高橋 亮

Project Assistant Professor, Graduate School of Engineering, The University of Tokyo
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Education

1. 2020.04–2023.03 **The University of Tokyo** The Graduate School of Engineering Department of Electrical Engineering and Information Systems
2. 2018.04–2020.03 **The University of Tokyo** The Graduate School of Information Science and Technology Department of Information and Communication Engineering
3. 2016.04–2018.03 **The University of Tokyo** The Faculty of Engineering Department of Information and Communication Engineering

Experience

1. 2026.01–2026.06 **Tsinghua Wireless Bioelectronics Group, Tsinghua Shenzhen International Graduate School at Tsinghua University** Visiting Researcher
2. 2025.10–2031.03 **JST CRONOS** Co-PI
3. 2025.10–2029.03 **JST PRESTO** Principal Investigator
4. 2022.12–2023.05 **Meta Inc., Reality Labs Research** Research Scientist Intern
5. 2022.04–2023.05 **Japan Society for the Promotion of Science (JSPS)** JSPS Research Fellow (DC2)
6. 2021.10–2025.03 **JST ACT-X** Principal Investigator
7. 2020.04–2020.05 **Shima Seiki Mfg., Ltd.** Knit design course (beginner)
8. 2019.04–2021.09 **JST ERATO Kawahara Universal Information Network Project** Research Assistant

Grants

1. 2026.06–2028.03 **Quasi-proximity data and power interface for smartring**
   Japan Society for the Promotion of Science (JSPS) KAKENHI Grant-in-Aid for Challenging Research (Exploratory)（26K23833） ・ Principal Investigator ・ JPY 4,900,000
2. 2026.04–2029.03 **Ultra-low-power ring-based AR controller for daily AR interface**
   Japan Society for the Promotion of Science (JSPS) KAKENHI Grant-in-Aid for Scientific Research (B)（26K02956） ・ Principal Investigator ・ JPY 14,100,000
3. 2025.10–2031.03 **Full-body wireless power and data networking**
   Japan Science and Technology Agency (JST) CRONOS (Innovative Science and Technology Initiative for Information and Communication) ・ Co-Investigator ・ JPY 42,160,000
4. 2025.10–2029.03 **Motion-robust body-scale distributed skin intelligence network**
   Japan Science and Technology Agency (JST) PRESTO（25146143） ・ Principal Investigator ・ JPY 40,000,000
5. 2025.04–2027.03 **Ubiquitous finger input interface using an ultra-low-power wireless ring mouse**
   The Asahi Glass Foundation Research Encouragement Grant (Physics and Information) ・ Principal Investigator ・ JPY 3,000,000

6. 2023.03–2024.03 **Body-scale wireless charging and sensing using a safe-to-body coil structure confining electromagnetic field near garment**
   Japan Society for the Promotion of Science (JSPS) KAKENHI Grant-in-Aid for JSPS Fellows （22KJ0810） ・ Principal Investigator ・ JPY 2,300,000
7. 2021.10–2025.03 **Building IoT systems with ubiquitous planar sensor arrays**
   JST ACT-X (Resilient Hardware) (+ acceleration phase) ・ Principal Investigator ・ JPY 19,834,000

Awards

1. 2026 **Best Paper Award** IEEE Robotics and Automation Letters (RA-L) 2025
2. 2026 **Funai Research Award** Funai Foundation for Information Technology
3. 2026 **Excellent Presentation Award A** IEEJ Sensors and Micromachines Society
4. 2025 **Outstanding Presentation Paper Award** 16th Symposium on Micro-Nano Science and Technology
5. 2025 **Encouragement Award** 42nd Sensor Symposium on Sensors, Micromachines and Applied Systems
6. 2025 **Outstanding Paper Award** IPSJ SIG-UBI, 87th Meeting
7. 2025 **Outstanding Paper Award** IPSJ SIG-UBI, 86th Meeting
8. 2024 **Outstanding Presentation Paper Award** 15th Symposium on Micro-Nano Science and Technology
9. 2024 **FSE Best Poster Presentation Award** 7th Event of Young Researchers Society for Flexible and Stretchable Electronics
10. 2023 **IPSJ Student Encouragement Award** 85th National Convention of IPSJ
11. 2023 **Telecom System Technology Student Award, Grand Prize** The Telecommunications Advancement Foundation Award
12. 2023 **IPSJ Yamashita SIG Research Award** Information Processing Society of Japan (IPSJ)
13. 2022 **Best Paper Award** 30th Workshop on Interactive Systems and Software (WISS 2022)
14. 2022 **Gaetano Borriello Outstanding Student Award Runner up** ACM Ubicomp 2022
15. 2022 **Best Paper Award** ACM CHI Conference on Human Factors in Computing Systems
16. 2021 **Outstanding Paper Award** IPSJ SIG-UBI
17. 2021 **CHI'20 @ CHI'21 People's Choice Best Demo Honorable Mention Award** Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)
18. 2020 **Honorable Mention Award** Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems Proceedings (ACM CHI)
19. 2020 **Interactive Presentation Award (PC and Audience Choice)** IPSJ Interaction

Publications

* First author † Corresponding author

Journal（10）

1. **Friction-Based Interconnection of Stretchable Liquid Metal-Based Coil with Rigid Capacitors for Mechanically Durable Wireless Powering Textiles**
   T. Sato*†, R. Takahashi*†, S. Watanabe, W. Yukita, T. Yokota, T. Someya, Y. Kawahara, E. Iwase, and W. Iwasaki
   ACS Applied Materials & Interfaces, Vol. 18, No. 37, pp. 51721-51730, 2026.
2. **Ultra-Low-Power Wireless Ring Mouse**
   R. Takahashi*†, Y. Li, M. Fukumoto, A. Noda, S. Ishida, T. Yokota, T. Someya, and Y. Kawahara
   Electronics and Communications in Japan, 146(7), pp.186-192, 2026.
3. **Near-field power and data networking via meandered e-textiles for continuous, large-scale BodyNet**
   R. Takahashi*† and Y. Kawahara
   IEEE Pervasive Computing, Vol. 25, No. 3, pp. 8-16, 2026.
4. **Knittable electronic textiles for intelligent soft wearables**
   R. Takahashi*, H. H. Chen, H. Mao, W. Yukita, T. Yokota, T. Someya, Y. Kawahara, Y. Luo†, and I. Wicaksono†
   npj Flex Electron, Published online, 2026.
5. **picoRing mouse:極低電力な指輪型無線マウス**
   高橋亮*†,Y. Li,福本雅朗,野田聡人,石田繁巳,横田知之,染谷隆夫,川原圭博
   電気論文誌E, 146巻,7号, 2026.
6. **Joint-repositionable Inner-wireless Planar Snake Robot**
   A. Kanada*†, R. Takahashi*, K. Hayashi, R. Hosaka, W. Yukita, Y. Nakashima, T. Yokota, T. Someya, M. Kamezaki, and Y. Kawahara
   IEEE Robotics and Automation Letters, Vol. 10, No. 5, pp. 4994-5001, 2025.
7. **Meander Coil++:継続的なウェアラブルコンピューティングのための衣類全面での安全で高効率な無線給電**
   高橋亮*†,雪田和歌子,横田知之,染谷隆夫,川原圭博
   コンピュータソフトウェア(サイバー増大号), Vol.40, No.3, pp. 113-122, 2023.
8. **Twin Meander Coil: Sensitive Readout of Battery-free On-body Wireless Sensors Using Body-scale Meander Coils**
   R. Takahashi*†, W. Yukita, T. Sasatani, T. Yokota, T. Someya, and Y. Kawahara
   Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT), Vol.5, No.4, Article No.179, 2021.
9. **Surface Routing for Wireless Power Transfer using 2-D Relay Resonator Arrays**
   M. Morita*, T. Sasatani, R. Takahashi, and Y. Kawahara†
   IEEE Access, Vol. 9, pp. 133102 - 133110, 2021.
10. **A Cuttable Wireless Power Transfer Sheet**
    R. Takahashi*†, T. Sasatani, F. Okuya, Y. Narusue, and Y. Kawahara
    Proc. of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT), Vol.2, No.4, Article No. 190, 2018.

Conference（13）

1. **Dense-Joint-Based Obstacle-Aided Locomotion with a Joint-Repositionable Snake Robot**
   K. Minomo*, R. Takahashi, K. Yasui, Y. Nakashima, M. Yamamoto, and A. Kanada
   2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), To appear, 2026.

2. **Machine-knittable, Magnetically-Plug-n-Play E-Textile Prototyping**
   Y. Li*, R. Takahashi* † , W. Yukita, I. Wicaksono, K. Matsutani, Y. Iwamoto, S. Lee, T. Yokota, T. Someya, and Y. Kawahara
   2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), To appear, 2026.

3. **picoRing dual: ultralow-power bimanual ring controllers for everyday AR**
   H. Yamamoto*, Y. Li, Y. Kawahara, and R. Takahashi †
   Proceedings of the 2026 ACM International Symposium on Wearable Computers (ISWC '26), To appear, 2026.

4. **3D-printing water-soluble channels filled with liquid metal for recyclable and cuttable wireless power sheet**
   T. Sato*, R. Takahashi, K. Yamagishi, T. Someya, M. Hashimoto, E. Iwase, Y. Kawahara, J. Kurumida, and W. Iwasaki
   2026 IEEE 39th International Conference on Micro Electro Mechanical Systems (MEMS), VII-b-2, 2026.

5. **Full-body WPT: wireless powering with meandered e-textiles**
   R. Takahashi* † , T. Sato, W. Yukita, T. Yokota, T. Someya, and Y. Kawahara
   Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion'25), pp. 529-533, 2025.

6. **Ultra-low-power ring-based wireless tinymouse**
   Y. Li*, M. Fukumoto, M. Kari, S. Ishida, A. Noda, T. Yokota, T. Someya, Y. Kawahara, and R. Takahashi †
   Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (ACM UIST), Article 22, 1-12, 2025.

7. **Plug-n-play e-knit: prototyping large-area e-textiles using machine-knitted magnetically-repositionable sensor networks**
   Y. Li*, R. Takahashi, W. Yukita, K. Matsutani, C. Caremel, Y. Iwamoto, S. Lee, T. Yokota, T. Someya, and Y. Kawahara
   Proceedings of the Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '25), Article No. 56, pp. 1-7, 2025.

8. **Friction Jointing of Distributed Rigid Capacitors to Stretchable Liquid Metal Coil for Full-Body Wireless Charging Clothing**
   T. Sato*, S. Watanabe, R. Takahashi, W. Yukita, T. Yokota, T. Someya, Y. Kawahara, E. Iwase, and J. Kurumida
   2025 IEEE 38th International Conference on Micro Electro Mechanical Systems (MEMS), pp. 181-184, 2025.

9. **picoRing: battery-free rings for subtle thumb-to-index input**
   R. Takahashi* † , E. Whitmire, R. Boldu, S. Ng, W. Kienzle, and H. Benko
   Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (UIST'24), Article No. 13, pp. 1-11, 2024.

10. **Meander Coil++: A Body-scale Wireless Power Transmission using Safe-to-body and Energy-efficient Transmitter Coil**
    R. Takahashi* † , W. Yukita, T. Yokota, T. Someya, and Y. Kawahara
    Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (ACM CHI'22), Article No.: 390, Pages 1-12, 2022.

11. **TelemetRing: A Batteryless and Wireless Ring-Shaped Keyboard using Passive Inductive Telemetry**
    R. Takahashi* † , M. Fukumoto, C. Han, T. Sasatani, Y. Narusue, and Y. Kawahara
    Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology (UIST'20), Pages 1161–1168, 2020.

12. **PneuModule: Using Inflatable Pin Arrays for Reconfigurable Physical Controls on Pressure-Sensitive Touch Surfaces**
    C. Han*, R. Takahashi, Y. Yahagi, and T. Naemura
    Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (ACM CHI'20), pp. 1-14, 2020.

13. **Topology Construction Protocol for Wireless Power Transfer System with a 2-D Relay Resonator Array**
    M. Morita*, T. Sasatani, R. Takahashi, and Y. Kawahara
    Proc. of IEEE International Conference on Consumer Electronics 2020 (ICCE), pp.1-5, 2020.

Workshop (5)

Organized

1. **Intelligent Soft Wearables: Bridging Disciplines Towards Soft Wearables for Everyone, Everywhere**
   C. Honnet*, T. C. Yu, R. Takahashi, T. Cheng, C. Zhang, Y. Kawahara, Y. Luo, I. Wicaksono, and J. Paradiso
   Proc. of ACM UbiComp 2026, To appear, 2026.

2. **Intelligent Soft Wearables**
   T. C. Yu*, C. Honnet, T. Cheng, R. Takahashi, B. Zhou, C. Zhang, J. A. Paradiso, Y. Kawahara, P. Lukowicz, J. Hester, Y. Luo, and I. Wicaksono
   Proc. of ACM UbiComp 2025, 1123-126, 2025.

Presented

3. **Collisionless Multi-Tag NFC Sensing for Indoor Activity Recognition Using Meander E-Textiles**
   S. Ido*, K. Tanaka*, R. Maeda*, R. Takahashi, A. Noda, and S. Ishida
   Proc. of ACM UbiComp 2026, Intelligent Soft Wearables Workshop, 2026.

4. **Flexible NFC wristband for ultralow-power ring-to-wristband interaction**
   H. Yamamoto*, Y. Li, Y. Kawahara, and R. Takahashi †
   Proc. of ACM UbiComp 2026, Intelligent Soft Wearables Workshop, 2026.

5. **Plug-n-play e-knit with LED-based sensor positioning**
   Y. Li*, R. Takahashi, I. Wicaksono, W. Yukita, Y. Iwamoto, S. Lee, T. Yokota, T. Someya, and Y. Kawahara
   Proc. of ACM UbiComp 2025, Intelligent Soft Wearables Workshop, 2025.

Demo (4)

1. **Body-scale NFC for wearables: human-centric body-scale NFC networking for ultra-low-power wearable devices (Demo of UTokyo Kawahara Lab 2025)**
   H. Yamamoto*, Y. Li*, W. Yukita, T. Yokota, T. Someya, R. Takahashi, and Y. Kawahara
   Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems, Article No. 825, pp. 1-6, 2026.

2. **Ultra-low-power ring-based wireless tinymouse**
   Y. Li*, M. Fukumoto, M. Kari, S. Ishida, A. Noda, T. Yokota, T. Someya, Y. Kawahara, and R. Takahashi †
   Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (ACM UIST), 2025.

3. **Demo of picoRing mouse: ultra-low-powered wireless mouse ring with ring-to-wristband coil-based impedance sensing**
   Y. Li*, M. Fukumoto, M. Kari, T. Yokota, T. Someya, Y. Kawahara, and R. Takahashi †
   Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), Article 707, pp. 1-5, 2025.

4. **Demo: Design of Cuttable Wireless Power Transfer Sheet**
   R. Takahashi*†, T. Sasatani, F. Okuya, Y. Narusue, and Y. Kawahara
   Adjunct Proc. of ACM UbiComp 2018, pp.456-459, 2018.

Poster（1）

1. **Body-scale wireless charging via meandered textile coil for continuous full-body electric skin**
   R. Takahashi*†, T. Sato, W. Yukita, T. Yokota, T. Someya, E. Iwase, and Y. Kawahara
   The 7th event of THE YOUNG RESEARCHERS SOCIETY FOR FLEXIBLE AND STRETCHABLE ELECTRONICS, 2024.

Domestic（29）

1. **NFCにおける短パルス間隔符号化に基づくアクセス制御**
   井戸智斗志*,石田繁巳,高橋亮
   超知性ネットワーキングに関する分野横断型研究会(RISING), 2026.

2. **メアンダコイル上の小型移動ロボットの走行中無線給電における給電効率を考慮した探索的経路生成手法**
   韓燦教*,高橋亮,横田知之,川原圭博
   ROBOMECH, 1P1-H09, 2026.

3. **picoRing NFC: 中距離NFCを用いた極低電力なワイヤレススマートリング**
   山本英昂*,高橋亮,川原圭博
   ROBOMECH 2026, 2A1-R04, 2026.

4. **機械編みによる人体スケールの NFC リーダ服**
   高橋亮*†,川原圭博
   ROBOMECH 2026, 2P1-O10, 2026.

5. **関節の高密度化がヘビ型ロボットの障害物利用推進に与える影響の調査**
   蓑毛響介*,高橋亮,安井浩太郎,山本元司,中島康貴,金田礼人
   ROBOMECH 2026, 2A1-I09, 2026.

6. **極低電力な指輪型無線マウス**
   高橋亮*†,Y. Li,福本雅朗,M. Kari,野田聡人,石田繁巳,横田知之,染谷隆夫,川原圭博
   「センサ・マイクロマシンと応用システム」シンポジウム, 第42回, 2025.

7. **液体金属を充填した3D印刷水溶性流路によるリサイクル可能なフレキシブル電子デバイス**
   佐藤峻*,高橋亮,山岸健人,染谷隆夫,橋本道尚,岩瀬英治,川原圭博,来見田淳也,岩﨑渉
   第16回マイクロ・ナノ工学シンポジウム, 12P2-PN-92, 2025.

8. **指輪-リストバンドコイル間の磁気結合を用いた超低電力な指輪型無線XRコントローラの設計**
   山本英昂*,李蘊韜,韓燦教,横田知之,染谷隆夫,川原圭博,高橋亮†
   信学ソ大2025, B-15-32, 2025.

9. **超低電力な指輪型無線マウス**
   高橋亮*†,Y. Li,福本雅朗,M. Kari,山本英昂,野田聡人,石田繁巳,横田知之,染谷隆夫,川原圭博
   情報処理学会 UBI研究会, 第87回, 2025.

10. **Roam beyond Battery: 長期的な屋内広範囲運用に向けた無線給電対応の小型移動ロボットネットワーク**
韓燦教*,高橋亮,野田聡人,川原圭博
ROBOMECH2025, 2A1-J10, 2025.

11. **移動関節ヘビ型ロボット:任意の体幹位置を能動屈曲可能なヘビ型ロボットによるセンサレス狭路推進制御の検討**
保坂琉介*,金田礼人,林渓人,中島康貴,高橋亮,雪田和歌子,横田知之,染谷隆夫,亀﨑允啓,川原圭博,佐藤峻,来見田淳也,岩瀬英治,山本元司
ROBOMECH2025, 2A1-R01, 2025.

12. **Design of full-body wireless charging-enabled textiles with liquid metal-based jointing**
T. Sato*, S. Watanabe, R. Takahashi, T. Yokota, T. Someya, Y. Kawahara, and E. Iwase
The 8th Event of the Researchers Society for Flexible and Stretchable Electronics, 2025.

13. **Full-body NFC: メアンダ形状の電子ニットによる全身電池レス無線センサネットワーク**
高橋亮*†,韓燦教,雪田和歌子,J. S. Ho,笹谷拓也,野田聡人,横田知之,染谷隆夫,川原圭博
UBI研究会, 第86回セッション2:IoT, 2025.

14. **強化学習による移動関節ヘビ型ロボットの移動パターン獲得**
黒田楓馬*,金田礼人,高橋亮,亀﨑允啓,川原圭博
情報処理学会 総合大会, 5J-05, 2025.

15. **picoRing: バッテリーレスな指輪型入力デバイス**
R. Takahashi*†, E. Whitmire, R. Boldu, S. Ng, W. Kienzle, and H. Benko
第32回インタラクティブシステムとソフトウェアに関するワークショップ(WISS2024), 国際学会招待発表枠, 2024.

16. **全身無線給電服に向けた分散コンデンサ付き液体金属メアンダ配線の検討**
佐藤峻*,渡邉慎人,高橋亮,川原圭博,岩瀬英治,来見田淳也
第15 回マイクロ・ナノ工学シンポジウム, マイクロナノシステム, 2024.

17. **医療デジタルツインの実現に向けた全身無線通信・給電服の設計**
高橋亮*†,野田聡人,韓燦教,横田知之,染谷隆夫,川原圭博
超知性ネットワーキングに関する分野横断型研究会(RISING), 41.0, 2024.

18. **医療デジタルツインの実現に向けた全身無線通信・給電服の設計**
高橋亮*†,野田聡人,韓燦教,横田知之,染谷隆夫,川原圭博
革新的無線通信技術に関する横断型研究会(MIKA2024), 80.0, 2024.

19. **自在関節索状ロボットのインナーワイヤレス化に向けた無線給電ソフトロボットスキンの検討**
黒田楓馬*,高橋亮,金田礼人,佐藤峻,亀﨑允啓,川原圭博
第42回日本ロボット学会学術講演会, 1C2-02, 2024.

20. **Meander Coil++:継続的なウェアラブルコンピューティングのための衣類全面での安全で高効率な無線給電**
高橋亮*†,雪田和歌子,横田知之,染谷隆夫,川原圭博
第30回インタラクティブシステムとソフトウェアに関するワークショップ(WISS 2022), pp.79-86, 2022.

21. **ユビキタスな面状センサアレイに向けた単一センサアレイによる多機能センシングの設計**
高橋亮*†,雪田和歌子,横田知之,染谷隆夫,川原圭博
信学総大 2022, B-15, 2022.

22. **衣服スケールのメアンダコイルによるバッテリレスセンサの高感度な無線読み取り手法**
高橋亮*†,雪田和歌子,笹谷拓也,横田知之,染谷隆夫,川原圭博
UBI研究会, 第71回, 2021.

23. **衣類全面での無線給電に向けた人体に安全な送電コイルの設計**
高橋亮*†,雪田和歌子,横田知之,染谷隆夫,川原圭博
信学ソ大 2021, B-15, 2021.

24. **バッテリレスなウェアラブルの無線センシングに向けた衣類型リーダコイルの設計**
    高橋亮*†,雪田和歌子,笹谷拓也,横田知之,染谷隆夫,川原圭博
    信学総大 2021, B-15-23, 2021.

25. **体内への電磁界暴露を抑制可能な無線給電・通信用テキスタイルコイルの設計**
    高橋亮*†,笹谷拓也,川原圭博
    信学ソ大 2020, B-15, 2020.

26. **マルチモード準静空洞共振器を用いた無線給電における中継器の構成についての検討**
    平井雄太*,笹谷拓也,高橋亮,川原圭博
    信学総大 2020, B-20-11, 2020.

27. **ワイヤレスな指輪型キーボードのバッテリレス化手法**
    高橋亮*†,韓燦教,笹谷拓也,成末義哲,福本雅朗,川原圭博,苗村健
    信学総大 2020, B-15-13, 2020.

28. **1MHz磁界共振結合方式を用いた無線給電プラレールの設計と製作**
    角谷和宣*,西澤勇輝,笹谷拓也,高橋亮,成末義哲,川原圭博
    信学技報, Vol. 118, No. 227, WPT2018-50, pp. 111-114, 2018.

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7. 2025 **Long-term full-body wearables powered by wireless electronic textiles** ADE@MIT, Online
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